

be your mobile phone, TVs, laptops and so on. Another isolated network will be there which is dedicated only for IoT devices. So these two are completely isolated from each other, and there is no connection between these two networks. However, the discovery and communication with the IoT devices can happen within the same range. Once the mesh data from the IoT devices are isolated within the network, unless you provide access to the personal WiFi network to somebody, and then they detect the device, only then the DDoS attack is possible. Till that point, to some extent you are safe, because you do not see that network at all. Let us say somebody goes on their phone, and looks for the WiFi networks around them that they can connect to, to guess the password or attempt to do a DDoS attack, they will not be able to see the network that the IoT devices are connected on at all. They will only be able to see the client based networks to provide connectivity to only the client devices. So that is one way that we are approaching the security aspect. But still, there are multiple ways in which somebody can try to attack.

Another weak protocol which is very vulnerable to attacks is Bluetooth Low Energy (BLE), which are even used for the locks on the doors. We are trying to work with the manufacturers and in our app we create a very secure password while you are discovering the BLE modules, and we create a PIN for it. Once you have created a PIN, it cannot be discovered again, unless you do a hardware reset from the inside of the door. This is possible only if you have access to the house. Only then will you be able to discover or see the lock. Any outsider will not be able to discover it if they are nearby. That way the locks are safeguarded. Now there are also cases where some part of the

network is encrypted or unencrypted. Most of the OEM's cloud platform based networking is unencrypted, only the bigger players invest in the encrypted methodology, where everything is decentralised, and the data is encrypted before being sent. In these devices, even the heartbeats are encrypted. With Smartden, the user will have discovered a specific set of devices from multiple manufacturers, and listed in our app. From here, we make sure to decentralise everything and only pass on encrypted data

to the cloud.. Even we will not be able to see that data, including the heartbeat or pings to know the status of the device. All the communications to our cloud servers are encrypted. Now there is another method that we are applying right now for our cloud communications to the manufacturers. If you are within the home, there is data that we send to our cloud, and another

set of data that we are sending to the manufacturers for the remote access. There are some OEMs we are working with, where we have our own servers for remote access, but for some of them we have to use the servers of the OEMs themselves, and their cloud platforms. So what we do is, we have found a method where that data is completely decentralized in an encrypted form before reaching the OEM cloud.

So from our side, we are trying to solve few of the security issues when it comes to the smarthome. There are more novel types of attacks, such as a multi stage attacks, that you can still do. Or you can apply machine learning and you can find out the password, or you can have Wire-shark around your home and keep sniffing the packets, but these are the problems that we are seeing since the inception of the internet. But these attacks are novel in nature. I cannot say if anyone can assure hundred percent

security, but there are some things that are in our hands, that we can do, and we are doing that.

d Are you compatible with Alexa?

Divyateja: In one way we can be considered as competition to Alexa. So we will not be using Alexa, but we might be able to integrate with Alexa speakers as well. Right now, we are not going in that direction. We will be coming up with our own voice assistant, within the phone itself.

d What kind of brands do you work with?

Divyateja: We are trying to partner with both local manufacturers and international ones. We are purely brand agnostic in that nature. What we are trying to do is work with the smarthome industry across the world to get the best smarthome products on the market, and to our platform. So we have tried and tested hundreds of devices, and we pick up the ones that are safe and secure. So we are a little picky in working with the OEMs.

d Can a consumer just use the app without opting for the services?

Divyateja: Yes. Smartden has two different aspects, one is through the services, and the other is to just download our apps from the store and use it. If you have lights from a specific brand, and that brand is supported by our app, then you can just start using it, you do not have to take our services.

d Where are the Smartden services available?

Niyati: We work all across the country. Our official presence is in Bengaluru, and our plans for launching in Delhi was delayed because of the second wave. We are also planning to launch in Mumbai by the end of the year. But let us say someone calls us from Lucknow, we are happy to provide our services.

Divyateja: We have done one project in a very rural part of Karnataka, so that way we are very comfortable working in any of the locations across the country. **d**



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